

Module 1

Introduction to System Administration

What is a system administrator?

Eh... that's Endministrator.



What is a system administrator?

- A person who is responsible for the upkeep, configuration, and reliable operation of computer systems, especially multi-user computers, such as servers.
- Often shorten as “sysadmin”

What is a system administrator?

- The system administrator seeks to ensure that the uptime, performance, resources, and security of the computers they manage meet the needs of the users, without exceeding a set budget when doing so.

10 minutes
Discussion !!

True or False:

With availability of public cloud services,
system administrators are no longer needed?

True or False:

With availability of public cloud services,
system [administration](#) are no longer needed?

True or False:

I want to work in the field of _____
(Software development, Cyber security, etc...)

so I **DO NOT** need to care
about system administration!?

What is system administration?

- System administration is the field of work in which someone manages one or more systems, be the hardware, software, and related processes.
- The goal is ensuring the systems are running efficiently and effectively.

Who is responsible for system administration then?

- System administration is typically done by information technology experts for or within an organization.
- **You're supposed to be an IT expert who is good at system administration!!**

IT systems

- Hardware
- System software
- Application software
- Management software
- Networking
- Users
- Processes
- Policies

Variants

- IT support
- Database administrator (DBA)
- Network administrator
- Web administrator
- Site Reliable Engineer
- Other job titles...

Well...



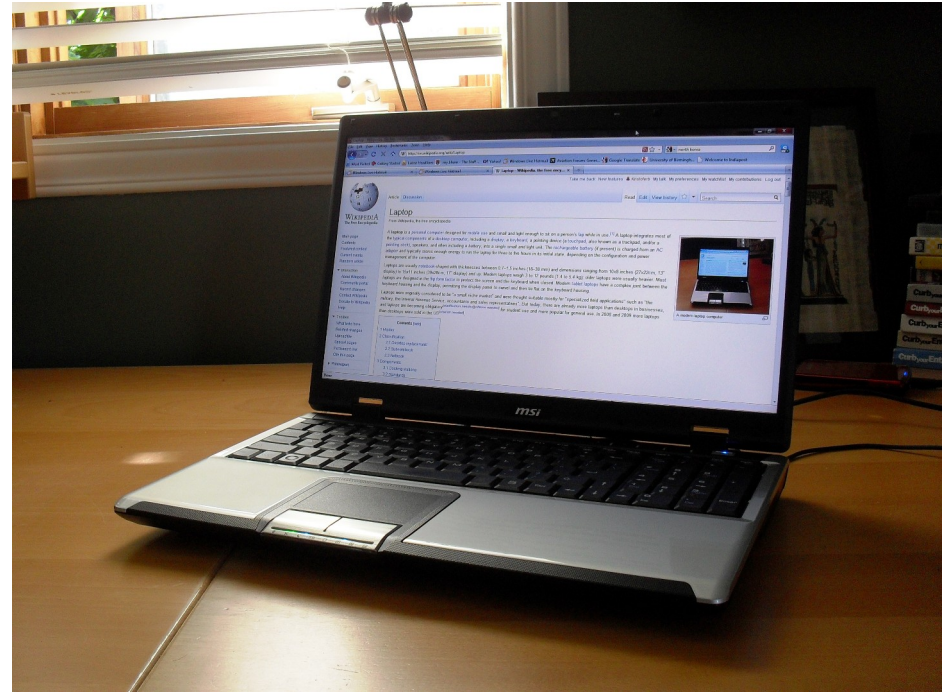
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Common systems for sysadmin

- Personal computers
- Workstations & Tower servers
- Rack servers
- Mainframes

Personal Computers



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Workstations and Tower servers



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Rack Servers



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Mainframes

- Over 70 percent of Fortune 500 companies still rely on mainframes despite the rise of cloud computing.*



* <https://planetmainframe.com/2022/12/relevance-of-mainframe/>

Sysadmin tasks

- Fault management
- Configuration management
- Security management
- Performance management
- Accounting management

Fault Management

- The detection of problems and faults on the system.
- Such faults should be properly logged, and if appropriate an alarm should be raised.
- This area is responsible for proper problem identification, determining the cause of the fault and ensuring the proper resolution of the problem.
- Management software operating in this area may attempt to correct faults on its own.

Configuration Management

- Keeping track of the system's configuration, both hardware and software.
- This area includes keeping track of what computers and networking infrastructure are on a network, and how they are interconnected.
- In addition, configuration management includes following what software versions each device is running, as well as the software configuration of each device.

Security Management

- Incorporating all aspects of authentication and access control, from the definition of access policies to the enforcement of those policies.
- Exception reports can be generated for events that fall out of the scope of the defined policies, and these reports can be used to alert administrators of the policy violation.

Performance Management

- Looking at the current and expected performance of the system.
- For example, elements of network performance that may be monitored include availability, response time, error rate, throughput and utilization.

Accounting Management

- covers two broad areas:
 - asset control
 - cost management
- Asset control refers to knowing what computers are on the network, who they belong to, who is using them, and perhaps where they are located.
- Cost management looks at what the costs of providing network services are and how they are paid for.

How should we learn sysadmin?

- Sysadmins often lack no precise job description.
- Best learn through practical experience.
- Learning all the technical topics related to sysadmin is too much.
- Within this course, we will learn the practice of sysadmin primarily through fundamentals of Linux system administration.